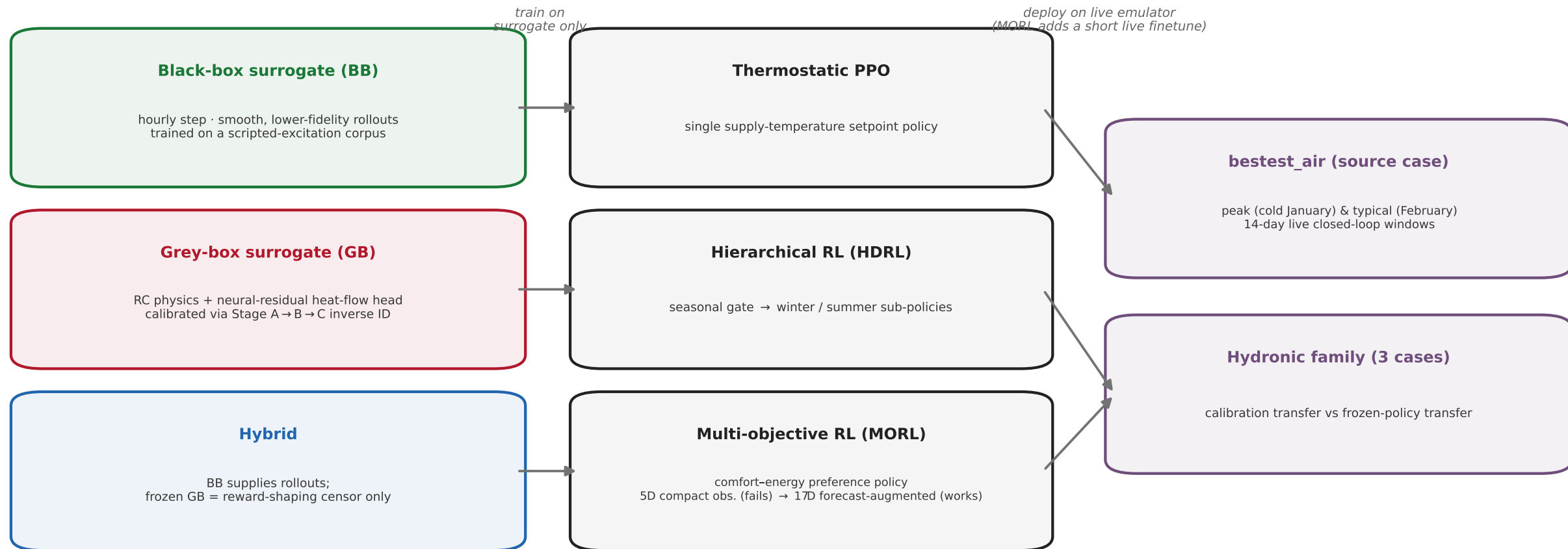


Surrogate training environments

Controller families (all PPO-based)

Live BOPTEST evaluation



Live metric: maintenance score m_s = time-in-violation + worst-case severity ($m_s > 1$ = unusable); also comfort-violation %, energy, 24 h rollout RMSE_T.

Schematic (no data).